CyberSentry Enterprise Assessment Report

Executive Vulnerability Assessment & Asset Intelligence Engine | RT-MSSU Capstone 2026-27 | Academic Research Lab

Target Network / CIDR:	google.com	**Assessment Date:**	2026-09-25 13:35:47 UTC
Scan Scope / Type:	UNIFIED	**Evaluation Body:**	RT-MSSU Security Evaluation Lab

Section 1: Executive Summary & Network Health Index

This security assessment document provides executive-grade threat intelligence and technical remediation guidance for **google.com**. The automated audit identified a total of **14 security findings**, including **2 Critical**, **1 High**, and **7 Medium** severity issues. The overall Enterprise Network Health Index is rated at **0/100**.

Network Health Index	Critical Findings (CVSS 9.0+)	High Risk (CVSS 7.0-8.9)	Medium Risk (CVSS 4.0-6.9)	Low Risk (CVSS <4.0)
0/100	2	1	7	4

Section 2: Asset Inventory Matrix

Discovered IP Address	Canonical Hostname	Inferred OS	Discovered Open Ports & Services
google.com	pu-in-f139.1e100.net	Generic / Unknown OS	80/http, 443/https

Section 3: Prioritized Vulnerability Register

CVE / Finding ID	Severity / CVSS	Vulnerability Title & Affected Target	CISA KEV / EPSS Status
CWE-319	HIGH CVSS 7.5	**Unencrypted Services** Port: Multiple | Target: google.com	Standard Threat EPSS: 38th percentile
CVE-2023-24488	CRITICAL CVSS 7.2	**Missing Content-Security-Policy (CSP)** Port: HTTP/HTTPS | Target: google.com	**Active KEV Listed** EPSS: 84th percentile
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CWE-319	MEDIUM CVSS 5.9	**Missing HTTP Strict Transport Security (HSTS)** Port: HTTP/HTTPS | Target: google.com	Standard Threat EPSS: 14th percentile
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CWE-319	MEDIUM CVSS 5.3	**Unencrypted HTTP Service (Port 80)** Port: 80 | Target: google.com	Standard Threat EPSS: 38th percentile
CWE-200	MEDIUM CVSS 4.3	**Server Header Software Disclosure** Port: HTTP/HTTPS | Target: google.com	Standard Threat EPSS: 8th percentile

CWE-16	**MEDIUM** CVSS 4.3	**Missing X-Content-Type-Options Header** Port: HTTP/HTTPS | Target: google.com	Standard Threat EPSS: 8th percentile
CWE-200	**MEDIUM** CVSS 4.3	**Server Header Software Disclosure** Port: HTTP/HTTPS | Target: google.com	Standard Threat EPSS: 8th percentile
CWE-16	**MEDIUM** CVSS 4.3	**Missing X-Content-Type-Options Header** Port: HTTP/HTTPS | Target: google.com	Standard Threat EPSS: 8th percentile
CWE-16	**LOW** CVSS 3.7	**Missing Permissions-Policy Header** Port: HTTP/HTTPS | Target: google.com	Standard Threat EPSS: 5th percentile
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CWE-116	**LOW** CVSS 2.5	**Missing Referrer-Policy Header** Port: HTTP/HTTPS | Target: google.com	Standard Threat EPSS: 5th percentile
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Section 4: Actionable Remediation Guide

System-by-system hardening recommendations and administrator execution playbooks:

1. Server Header Software Disclosure

Impact: Provides attackers with exact software version numbers to tailor targeted exploit payloads.

Action Plan:
• Disable server signature tokens in Nginx (`server_tokens off;`) or Apache (`ServerTokens Prod`).
• Strip `X-Powered-By` header in application middleware.
• Restart web server.

Configuration / CLI Fix:

```
# Nginx:
server_tokens off;

# Apache (in httpd.conf / security.conf):
ServerTokens Prod
ServerSignature Off
```

2. Missing Content-Security-Policy (CSP)

Impact: Allows attackers to execute malicious JavaScript in user sessions, hijacking session cookies and performing unauthorized actions.

Action Plan:
• Implement contextual HTML and JavaScript output encoding before rendering user input.
• Enforce Content Security Policy (CSP) with `script-src 'self'`.
• Set `HttpOnly` and `Secure` flags on all session cookies.

Configuration / CLI Fix:

```
# Nginx: Set Content Security Policy Header
add_header Content-Security-Policy "default-src 'self'; script-src 'self';" always;
add_header Set-Cookie "Path=/; Secure; HttpOnly; SameSite=Strict";
```

3. Missing HTTP Strict Transport Security (HSTS)

Impact: Allows attackers in a Man-in-the-Middle position to strip TLS and downgrade HTTPS traffic to cleartext HTTP.

Action Plan:

• Add `Strict-Transport-Security` header with `max-age=31536000; includeSubDomains`.
• Consider submitting the domain to the Chrome HSTS Preload List.

Configuration / CLI Fix:

```
# Nginx:
add_header Strict-Transport-Security "max-age=31536000; includeSubDomains; preload" always;

# Apache:
Header always set Strict-Transport-Security "max-age=31536000; includeSubDomains"
```

4. Missing X-Content-Type-Options Header

Impact: Browsers may perform MIME sniffing on untrusted user uploads, interpreting them as executable scripts.

Action Plan:

• Add `X-Content-Type-Options: nosniff` header to all web responses.

Configuration / CLI Fix:

```
# Nginx:
add_header X-Content-Type-Options "nosniff" always;

# Apache:
Header always set X-Content-Type-Options "nosniff"
```

5. Missing Permissions-Policy Header

Impact: Fails to restrict client browser APIs such as camera, microphone, geolocation, and payment delegation.

Action Plan:

• Implement modern `Permissions-Policy` header in web server configuration.

Configuration / CLI Fix:

```
# Nginx:
add_header Permissions-Policy "camera=(), microphone=(), geolocation=(), payment=()" always;
```